

A

One dictionary, five instantiations

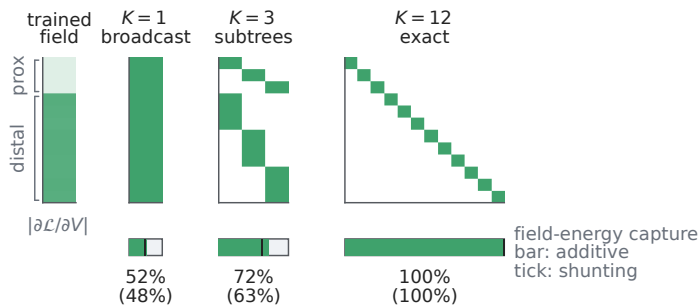


addresses A_u ($K = 4$)
on one arbor

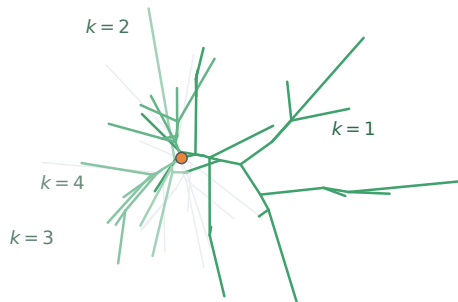
- A_u address matrix
delivery model (Figs 1, 3)
- Haar route basis
operator screens (Fig. 4)
- $\Phi^{(K)}$ context routes
eight-context task (Fig. 6)
- A_K anatomical subtrees
reconstructed arbors (Fig. 7)
- fitted subtree routes
measured responses (Fig. 9)

B

Dictionary gallery on a [3,3] arbor

**C**

Anatomical routes (MICrONS)

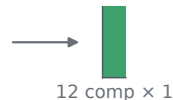


4 of 9 depth-1 routes cover 58/77 segments

D

Which dictionary a task demands

image classes
MNIST ladder (Fig. 3)



one coefficient
per neuron suffices

conflicting branches
 $\chi > \chi_c$ (Fig. 5)



branch selectors
become necessary

nested contexts
eight-context task (Fig. 6)



subtrees win in a
 $K=4$ window